



**UNIVERSITY
OF LONDON**

MSc Data Science

DSM010 Big Data Analysis

Midterm coursework assignment: Apr 2026 to Sept 2026 study session

- Please note you are permitted to upload your coursework in the final submission area as many times as you like before the deadline. You will receive a similarity/originality score which represents what the Turnitin system identifies as work similar to another source. The originality score can take over 24 hours to generate, especially at busy times e.g. submission deadline.
- If you upload the wrong version of your coursework, you are able to upload the correct version of your coursework via the same submission area. You simply need to click on the 'submit paper' button again and submit your new version before the deadline.

In doing so, this will delete the previous version which you submitted, and your new updated version will replace it. Therefore, your Turnitin similarity score should not be affected. If there is a change in your Turnitin similarity score, it will be due to any changes you may have made to your coursework.

- Please note, when the due date is reached, the version you have submitted last, will be considered as your final submission and it will be the version that is marked.
- **Once the due date has passed, it will not be possible for you to upload a different version of your assessment. Therefore, you must ensure you have submitted the correct version of your assessment which you wish to be marked, by the due date.**

Coursework description

In this coursework, you will tackle advanced problems using the MapReduce computational model and Mahout on a Hadoop cluster. The midterm coursework assignment is worth 30 marks and constitutes 30% of your total mark for the module.

Question 1 – Distributed Weather Data Analysis and Predictive Modelling

Data Source: Utilise the hourly weather data from the NCDC for April, May, June, and July 2007, accessible in the 'Data Sets' folder under the "Coursework submission" tab on the module VLE.

Objective: Employ the MapReduce framework to perform distributed statistical analysis and predictive modelling on weather variables without using ready-made statistical packages.

Implementation requirement for Question 1:

For all parts of Question 1, you must provide clear pseudocode and Python implementation of the mapper and reducer functions. Your code should include sufficient comments so that tutors can understand and reproduce your results. You should not use ready-made statistical packages for computing the required statistics or regression coefficients.

Deliverables:

1. Calculate the daily minimum, maximum, mean, and standard deviation of "Dry Bulb Temp" for your selected month. In addition, compute the daily average and maximum "Wind Speed" across all weather stations.
2. Compute the covariance and correlation between "Dry Bulb Temp" and two other weather variables of your choice, such as "Relative Humidity", "Dew Point Temp", "Wet Bulb Temp", or "Wind Speed". Interpret the strength and direction of these relationships.
3. Develop MapReduce jobs to calculate station-level summary statistics for "Dry Bulb Temp" and "Wind Speed", including count, mean, **population variance**, standard deviation, minimum, and maximum. For consistency across submissions, use population variance unless you explicitly state and justify the use of sample variance.
4. Construct MapReduce jobs to fit a simple linear regression model that predicts "Dry Bulb Temp" using one selected explanatory variable, such as "Dew Point Temp" or "Relative Humidity". Develop mapper and reducer functions to calculate the regression coefficients and evaluate the model using an error measure such as Mean Squared Error.
5. Extend the regression analysis by using two explanatory weather variables to predict "Dry Bulb Temp". Explain how the required summary quantities can be computed in a distributed manner using

MapReduce. Compare the performance of the one-variable and two-variable models.

6. The mapper and reducer pseudocode and Python code must be provided for all relevant computations in Question 1.

Statistical Computation Guide:

For this coursework, unless otherwise stated, use **population variance**:

$$\text{variance} = 1/N \times (\sum x_i^2 - N \times \bar{x}^2)$$

For simple linear regression:

Mapper outputs key-value pairs:

key = 'constant_identifier', value = (x, y)

Reducer computes:

$\sum x$, $\sum y$, $\sum x^2$, $\sum xy$, and n

The slope and intercept are then calculated as follows:

$$\text{Slope (m)} = (n \times \sum xy - \sum x \times \sum y) / (n \times \sum x^2 - (\sum x)^2)$$

$$\text{Intercept (b)} = (\sum y - m \times \sum x) / n$$

For Mean Squared Error, use:

$$\text{MSE} = 1/N \times \sum (y_i - \hat{y}_i)^2$$

Question 2 – Document Clustering using Apache Mahout and K-Means

For this question, you should use Apache Mahout to perform K-Means clustering on a collection of text documents. You may use the text dataset provided with Topic 4, or you may choose your own text dataset. If you use your own dataset, you must provide a link to the dataset in your report and briefly explain why it is suitable for clustering.

In text clustering, documents are represented using the terms that appear in them. Since machine learning algorithms cannot directly process raw text, the documents must first be converted into numerical feature vectors. Apache Mahout supports this process by transforming raw text into sequence files and then into vector representations that can be used for clustering.

Your task is to apply K-Means clustering using Apache Mahout and evaluate the clustering results.

You should design and conduct a Mahout-based K-Means clustering experiment on a suitable text dataset. Your work should include:

1. A brief technical explanation of your preprocessing workflow, including conversion from raw text to sequence files and from sequence files to sparse numerical vectors.
2. A methodological justification of your experimental design, including your choice of dataset, feature representation, values of K , and distance measures.

3. K-Means clustering experiments for at least three different values of K and at least two distance measures, such as Euclidean, Manhattan, or Cosine distance.
4. A table comparing clustering performance across different values of K and distance measures, using an appropriate measure such as average distance to centroid.
5. An elbow graph or equivalent visual/quantitative justification for the selected value of K .
6. An interpretation of the final clusters, including the main themes/topics and whether the clusters are meaningful.
7. A critical discussion of the limitations of K-Means for text data, including sensitivity to K , dependence on feature representation, distance-measure choice, and interpretability.

Your report should include the experimental results, a short explanation of the workflow, and a discussion of the quality of the clusters obtained.

Question 3 – Use a classifier to understand text similarity

For this question you are expected to use a transformer model in Hadoop to understand similarity between two pieces of text.

Transformers have proven to be foundational for text understanding and extraction and have become the default tool for such tasks. They are also heavily utilised in chat-based systems.

The dataset to use is `it_tickets.csv` and it contains two columns of IT tickets that have been extracted from a database and matched based on an error. Please compute similarity between the tickets and output a True or False value if the tickets are similar or not. You must define and justify the threshold used to convert similarity scores into True or False labels. Your justification should briefly discuss the trade-off between precision and recall. For example, a higher threshold may reduce false positives but may miss genuinely similar ticket pairs; a lower threshold may identify more similar pairs but may increase false positives.

For this task, you may use models from Hugging Face or models from Sentence Transformers.

If you are using Hugging Face based models, please use the following two lines at the top of your script:

```
import os  
os.environ['HF_HOME'] = '.cache'
```

If you are using Sentence Transformers based models, please use the following parameter when instantiating a SentenceTransformer:

```
cache_folder='.'  
i.e. SentenceTransformer(<MODEL NAME>, cache_folder='.')
```

Note: *Please consider executing models in the reducer, as mappers are intended to perform only minimal processing on the data.*

You need to include the following in your coursework submission:

- (a) For Question 1 submit the pseudo code and Python code for the mappers and reducers implementation for all of the descriptive statistics, along with some comments so that a layperson can implement. Anyone should be able to run your code and reproduce your results with the instructions that you have provided.
- (b) For Question 2, write a brief summary on the impact of parameter changes on the performance of K-Means algorithm. For example, you may: 1) compare different distance measures in K-Means algorithm; discuss the merits and demerits and 2) present a table that shows the performance of K-Means algorithm for different K values.
- (c) For Question 3, submit the classifier (that has a mapper and a reducer) and the results.
- (d) Submit a report on the experiments. This report will be a detailed explanation (max 2,000 words, excluding code and references) of what you explored, the results you obtained, and some discussion points on the limitations of MapReduce methodology and Hadoop's MapReduce computing engine.
- (e) Submission format:
 - PDF report of maximum 2,000 words, excluding code, references, tables, and appendices. Please use the following file-naming conventions: **StudentNumber_DSM010_CW1.pdf**
 - one ZIP file, named **StudentNumber_DSM010_CW1_implementation.zip**, containing:
 - Python mapper and reducer files for Question 1 and Question 3.
 - Any scripts or command files used for Mahout experiments in Question 2.
 - Output files or selected result extracts from HDFS.
 - A short README.txt explaining how to run the code and where each result can be found.

Evidence of Hadoop/HDFS execution:

For each task, your report should include appropriate evidence that the experiments were run on the Hadoop cluster. This may include screenshots of Hadoop/Mahout jobs running, selected job logs, HDFS output extracts, command-line screenshots, formatted tables of numerical results, and brief interpretations of what the results mean. Screenshots should support the results but should not replace properly formatted tables and discussion.

Credit will be given for:

- The depth and breadth of your investigation.
- The technical skills you demonstrate in your write-up.
- Good use of the Hadoop cluster.
- Critical evaluation of your work.

Activity	Marks
Question 1: MapReduce weather statistics and predictive modelling	10
Q1 descriptive statistics, covariance/correlation, station-level summaries, mapper/reducer design, implementation, and results	6
Q1 simple and two-variable regression, distributed computation of coefficients, MSE comparison, and interpretation	4
Question 2: Mahout K-Means document clustering	12
Mahout workflow, vectorisation, K-Means implementation, and use of at least two distance measures, e.g. Euclidean and Manhattan or Cosine	3
Study of the impact of different (K) values, including results table	2
Distance-measure comparison and interpretation	2
Elbow graph or equivalent justification for selected (K)	2
Critical discussion of MapReduce/Mahout methodology and limitations of K-Means for text clustering	3
Question 3: Transformer-based text similarity/classification	4
Mapper/reducer classifier or similarity pipeline, threshold justification, and results	4
Overall report quality	4
Clarity, reproducibility, evidence of Hadoop/HDFS execution, structure, presentation, and critical evaluation	4
Total	30

[END OF COURSEWORK ASSIGNMENT 1]